

Ethan

PUYAUBREAU

HPC / DevOps Engineer

C++ · EDF R&D · Oak Ridge (USA)

Available January 2027

CONTACT

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 github.com/ethan-puyaubreau

TECHNICAL SKILLS

HPC & Scientific Computing

C++17/20 · CUDA · Kokkos · CMake

Performance Profiling · MPI · OpenMP

Infrastructure & Platform

Kubernetes (K3s) · Docker · Ceph

Proxmox VE · pfSense · Ansible · Traefik

Linux / Debian

CI/CD & DevOps

GitLab CI/CD · Jenkins · GitHub Actions

Debian Packaging · Git

Languages

C++ · Python · TypeScript · Java · Bash

Learning: Rust · Go

Data & Backend

PostgreSQL · MySQL · JSON/XML/XSD

Docs

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LANGUAGES

FrenchNative

EnglishC1 — TOEIC 970/990

SpanishB1

CERTIFICATIONS

TOEIC C1 — 970/990

ETS · October 2025

INTERESTS

Flight Simulation (DCS World, KSP, VR)

3D & VFX · Music · Self-hosting

PROFILE

C++ / HPC software engineer trained through a work-study program: three years at EDF R&D on the platform that simulates the group’s nuclear reactor cores, including a research stay at Oak Ridge National Laboratory (USA) during the apprenticeship. Dual background: high-performance computing (profiling, GPU, Kokkos) and infrastructure/DevOps (Kubernetes, CI/CD, Debian packaging), sustained by a personal multi-server infrastructure operated for five years. **Available for full-time positions from January 2027.**

WORK EXPERIENCE

Software Architecture & Scientific Computing Engineer (work-study)

EDF Lab Paris-Saclay · Palaiseau, France

Sept. 2023 – Sept. 2026

- Developed the prototype of the new compute-core architecture of the nuclear simulation platform: results rigorously identical to the reference, **bit-for-bit**, up to **12 % faster** and **40 % lower peak memory** (neutronic solvers, C++).
- Built two performance diagnostic tools for this **500k-line** platform developed by **30+ engineers**: memory consumption analysis and compute-time measurement (C++, Python bindings); they were used to validate the prototype.
- Automated the test and delivery chain: cluster execution, result checking and storage, release packaging (Jenkins, GitLab CI/CD, XML/XSD, PostgreSQL, Scibian).
- Teamwork: code reviews given and received, leading technical meetings, user support (e.g. a blocking memory blow-up the team had chased for days, identified with the profiling tool).
- Written communication: **5 internal technical notes**, **3 apprenticeship reports** with annual defenses.

C++17 · CUDA · Python · PyBind11 · CMake · PostgreSQL · GitLab CI/CD · Jenkins · Linux Scibian

Graduate Research Fellow — GRO Program

Oak Ridge National Laboratory · Oak Ridge, TN, USA

June – Aug. 2025

- Research stay during the apprenticeship: developed two tools measuring GPU energy consumption for Kokkos, a major scientific computing framework, tested on **Frontier**, then the world’s most powerful supercomputer (**1.2 exaflops**).
- Tools compatible with NVIDIA and AMD GPUs (NVML, ROCm-SMI), measurement error < **5 %**, validated on multi-node simulations (LAMMPS, MPI).
- Two posters presented** (ORNL internal session, Smoky Mountains Conference 2025); invited to present at Supercomputing (SC25), declined due to the apprenticeship schedule.
- Open source contributions: **9 PRs to kokkos-tools and LAMMPS**, **3 merged**, ongoing collaboration with maintainers.
- Findings presented back at EDF Lab to the group and department group leaders.

C++17 · Kokkos · CUDA · MPI · SLURM · HPC · Open source

Software Engineering Intern

EDF Lab Paris-Saclay · Palaiseau, France

Apr. – June 2022

Fault trees and bow-tie models in Python for nuclear power plant risk analysis; functional prototype intended for internal use.

EDUCATION

Master of Engineering (M.Eng.) — Computer Science & Applied Math

Polytech Paris-Saclay · Université Paris-Saclay, France

Sept. 2023 – Sept. 2026

Work-study program (3 years in industry). CTI-accredited, equivalent to a Bologna-process M.Eng./M.Sc. Scientific computing · HPC · Parallel programming · Software architecture · Networks & security · ML.

Bachelor of Science (B.S.) — Sciences and Technology

Institut Villebon Georges Charpak · Orsay, France

Sept. 2019 – June 2023

PROJECTS

Personal Multi-Server Infrastructure

github.com/ethan-puyaubreau

June 2020 – Present

Designed and operated end to end, solo, for five years: Proxmox VE virtualization, Kubernetes (K3s) orchestration, Ceph distributed storage, segmented network (pfSense, VLANs), monitoring and CI/CD (GitLab, Ansible). **Around twenty services running continuously** for ~60 regular users.

Proxmox VE · Kubernetes · Ceph · Docker · pfSense · Ansible · GitLab CI/CD

Vireli — Carbon Footprint PWA (team lead)

Polytech Paris-Saclay · industry partnership with Viveris

2025 – 2026

Led a team of six (554 cumulative hours): overall architecture, backend (Hono, Bun, Prisma), deployment infrastructure (Docker, Gitea CI, Coolify, PostgreSQL, Redis). **18/18 functional requirements delivered** and validated.

TypeScript · PostgreSQL · Docker · CI/CD · Team lead

Large-Scale Online Event Organization

2021 – Present

Multiplayer events with **150 to 180+ participants**, recognized by specialized press: coordination of a 4–5 person staff, live technical monitoring (Grafana, Prometheus), systematic debriefs.

Coordination · Monitoring · Communication